

INTERNNOVA – DATA ANALYTICS INTERNSHIP

Python Fundamentals for Data Analytics

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Batch: Batch-03

Domain: Data Analytics

Internship Duration: 01 August 2026 – 11 September 2026

Task 1: Python Basics

Brief Explanation:

This program takes the student's name, college name, and branch as input and displays the entered information in a formatted output.

Python Source Code:

```
print("welcome to the python program")
name = input("Please enter your name: ")
collage = input("Please enter your college: ")
branch = input("Please enter your branch: ")

print('/n-----student details-----')
print('name:', name)
print('college:', collage)
print('branch:', branch)
```

Output:

```
web disinging/D1 sample/task1.py"
welcome to the python program
Please enter your name: adithyan
Please enter your college: sreenarayana collage vadakara
Please enter your branch: bca
/n-----student details-----
name: adithyan
college: sreenarayana collage vadakara
branch: bca
PS C:\Users\ASUS\AppData\Local\Programs\Microsoft VS Code>
```

Task 2: Variables & Data Types

Brief Explanation

This program demonstrates different Python data types such as integer, float, string, and boolean. It also uses the `type()` function to display the data type of each variable.

Code:

```
age = 21
height = 160.5
name = "Adithyan"
is_student = True

print("Age:", age)
print("Data Type:", type(age))

print("Height:", height)
print("Data Type:", type(height))

print("Name:", name)
print("Data Type:", type(name))

print("Student:", is_student)
print("Data Type:", type(is_student))
```

Output:

```
web_disinging/D1 sample/task2.py
Age: 21
Data Type: <class 'int'>
Height: 160.5
Data Type: <class 'float'>
Name: Adithyan
Data Type: <class 'str'>
Student: True
Data Type: <class 'bool'>
PS C:\Users\ASUS\AppData\Local\Programs\Microsoft
```

TASK 3 – Operators

Brief Explanation

This program creates a simple calculator that takes two numbers as input and performs addition, subtraction, multiplication, division, and modulus operations.

Code:

```
print("===== Simple Calculator =====")

num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

addition = num1 + num2
subtraction = num1 - num2
```

```

multiplication = num1 * num2
division = num1 / num2
modulus = num1 % num2

print("\n===== Results =====")

print("Addition:", addition)
print("Subtraction:", subtraction)
print("Multiplication:", multiplication)
print("Division:", division)
print("Modulus:", modulus)

```

Output:

```

Enter the second number: 5

===== Results =====
Addition: 9.0
Subtraction: -1.0
Multiplication: 20.0
Division: 0.8
Modulus: 4.0
PS C:\Users\ASUS\AppData\Local\Programs\Microsoft VS Code>

```

TASK 4 – Conditional Statements

Brief Explanation

This program takes marks as input and displays the appropriate grade using if, elif, and else conditional statements.

Code:

```

print('====grade calculator====')
marks=int(input('enter the marks:'))
if marks>=90:
    print('grade: A')
elif marks>=75:
    print("grade: B")
elif marks>=60:
    print("grade: C")
else:
    print('grade: fail')

```

Output:

```

web disinging/D1 sample/task4.py"
====grade calculator====
enter the marks:67
grade: C

```

TASK 5 – Loops

Brief Explanation

This program demonstrates the use of for and while loops. It prints numbers from 1 to 20, generates the multiplication table of a user-entered number, and prints even numbers from 1 to 50.

Code:

```
print("Numbers from 1 to 20")
for i in range(1,21):
    print(i)

number =int(input('Enter a number:'))
print('\nMultiplication table for')
for i in range(1,11):
    print(number,'x',i,'=',number*i)

print('Even numbers from 1 to 50')
num=2
while num<=50:
    print(num)
    num=num+2
```

Output:

```
web dising/D1 sample/task5.py"
36
38
40
42
44
46
48
50
```

TASK 6 - Functions

Brief Explanation

This program demonstrates user-defined functions in Python. It contains a function to calculate the square of a number and another function to calculate the average of three numbers. Both functions are called using user input.

Code:

```
# Task 6 - Functions

# Function to calculate square
def square(number):
    return number * number

# Function to calculate average
def average(num1, num2, num3):
    return (num1 + num2 + num3) / 3

print("==== Functions Program =====")

# Input for square
number = int(input("Enter a number: "))
```

```

# Call square function
result = square(number)

print("Square =", result)

print()

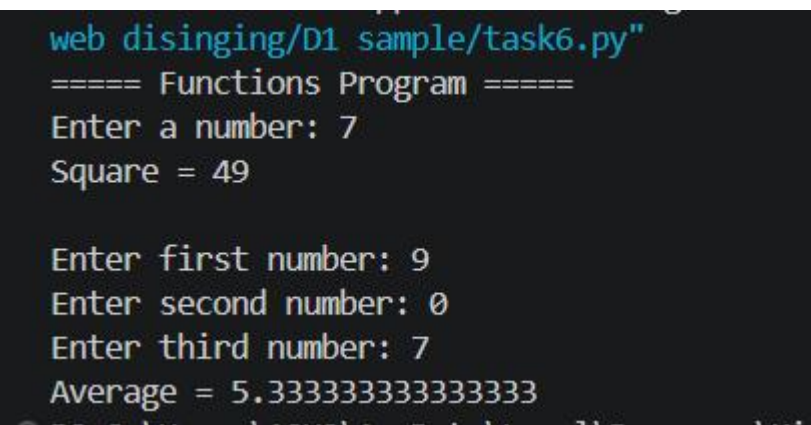
# Input for average
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

# Call average function
avg = average(a, b, c)

print("Average =", avg)

```

Output:



```

web disinging/D1 sample/task6.py"
===== Functions Program =====
Enter a number: 7
Square = 49

Enter first number: 9
Enter second number: 0
Enter third number: 7
Average = 5.333333333333333

```

TASK 7 - Strings & Collections

Brief Explanation

This program demonstrates Python string operations and collection data structures. It uses string methods such as `upper()`, `lower()`, `replace()`, and `find()`, along with list, tuple, dictionary, and set operations.

Code:

```

# =====
# Task 7 - Strings & Collections
# =====

print("===== Task 7 : Strings & Collections =====")

# -----
# STRING OPERATIONS
# -----

text = input("Enter a sentence: ")

print("\n--- String Operations ---")

print("Original String :", text)

print("Uppercase :", text.upper())

print("Lowercase :", text.lower())

print("Replace Python with Data Analytics :")

```

```

print(text.replace("Python", "Data Analytics"))

word = input("Enter a word to find: ")

position = text.find(word)

print("Position of the word :", position)

# -----
# LIST OPERATIONS
# -----

print("\n--- List Operations ---")

numbers = []

for i in range(5):
    num = int(input(f"Enter Number {i+1}: "))
    numbers.append(num)

print("Original List :", numbers)

remove_number = int(input("Enter a number to remove: "))

if remove_number in numbers:
    numbers.remove(remove_number)
    print("After Remove :", numbers)
else:
    print("Number not found in the list.")

numbers.sort()

print("Sorted List :", numbers)

# -----
# TUPLE
# -----

print("\n--- Tuple ---")

student = ("Adithyan", 21, "BCA")

print("Tuple :", student)

print("First Element :", student[0])

print("Second Element :", student[1])

print("Third Element :", student[2])

# -----
# DICTIONARY
# -----

print("\n--- Dictionary ---")

student_info = {
    "Name": "Adithyan",
    "Age": 21,
    "Branch": "BCA",
    "College": "University of Calicut"
}

print(student_info)

print("Student Name :", student_info["Name"])

```

```
print("Branch :", student_info["Branch"])
```

```
# -----  
# SET  
# -----
```

```
print("\n--- Set ---")
```

```
colors = {"Red", "Blue", "Green"}
```

```
print("Original Set :", colors)
```

```
colors.add("Yellow")
```

```
print("After Add :", colors)
```

```
colors.remove("Blue")
```

```
print("After Remove :", colors)
```

Output:

```
eb dising/D1 sample/task7.py"
===== Task 7 : Strings & Collections =====
Enter a sentence: adhi

--- String Operations ---
Original String : adhi
Uppercase : ADHI
Lowercase : adhi
Replace Python with Data Analytics :
adhi
Enter a word to find: i
Position of the word : 3

--- List Operations ---
Enter Number 1: 2
Enter Number 2: 6
Enter Number 3: 7
Enter Number 4: 8
Enter Number 5: 9
Original List : [2, 6, 7, 8, 9]
Enter a number to remove: 6
After Remove : [2, 7, 8, 9]
Sorted List : [2, 7, 8, 9]

--- Tuple ---
Tuple : ('Adithyan', 21, 'BCA')
First Element : Adithyan
Second Element : 21
Third Element : BCA

--- Dictionary ---
{'Name': 'Adithyan', 'Age': 21, 'Branch': 'BCA', 'College': 'University of Calicut'}
Student Name : Adithyan
Branch : BCA

--- Set ---
Original Set : {'Green', 'Red', 'Blue'}
After Add : {'Green', 'Yellow', 'Red', 'Blue'}
After Remove : {'Green', 'Yellow', 'Red'}
```

TASK 8 - Basic File Handling

Brief Explanation

This program demonstrates basic file handling in Python. It creates a text file, writes the student's introduction into the file, reads the file contents, and displays them on the screen.

Code:

```
> Users > ASUS > OneDrive > Desktop > web disinging > D1 sample > task8.py >
1 print("====Basic file Handling====")
2 file = open('introduction.txt', 'w')
3 file.write('name: Adithyan\n')
4 file.write('age: 21\n')
5 file.write('branch: BCA\n')
6 file.write('college: University of Calicut\n')
7 file.write('domain: data analytics\n')
8 file.close()
9 print("File created successfully!")
10 file = open('introduction.txt', 'r')
11 content = file.read()
12 print('\n====File Content====')
13 print(content)
14 file.close()
```

```
PS C:\Users\ASUS\AppData\Local\Programs\Microsoft VS Code> python eb disinging/D1 sample/task8.py"
=====Basic file Handling=====
File created successfully!

=====File Content=====
name: Adithyan
age: 21
branch: BCA
college: University of Calicut
domain: data analytics

PS C:\Users\ASUS\AppData\Local\Programs\Microsoft VS Code>
```

Brief Explanation

This project develops a Student Record Management System using Python. It allows the user to add student details, display student records, search for a student by name, and delete a student record. Lists and dictionaries are used for storing student information.

Code:

```
# =====  
# Task 9 - Student Record Management System  
# =====  
  
students = []
```

while True:

```
print("\n==== Student Record Management System ====")
print("1. Add Student")
print("2. Display Students")
print("3. Search Student")
print("4. Delete Student")
print("5. Exit")
```

```
choice = input("Enter your choice: ")
```

```
# -----
# Add Student
# -----
```

```
if choice == "1":
```

```
    name = input("Enter Student Name: ")
    age = input("Enter Age: ")
    branch = input("Enter Branch: ")
```

```
    student = {
        "Name": name,
        "Age": age,
        "Branch": branch
    }
```

```
    students.append(student)
```

```
    print("Student Added Successfully!")
```

```
# -----
# Display Students
# -----
```

```
elif choice == "2":
```

```
    if len(students) == 0:
        print("No Student Records Found.")
```

```
    else:
```

```
        print("\nStudent Records")
```

```
        for student in students:
            print(student)
```

```
# -----
# Search Student
# -----
```

```
elif choice == "3":
```

```
    search = input("Enter Student Name: ")
```

```
    found = False
```

```
    for student in students:
```

```
        if student["Name"].lower() == search.lower():
```

```
            print("\nStudent Found")
            print(student)
```

```
            found = True
```

```
    if found == False:
```

```
        print("Student Not Found.")
```

```
# -----  
# Delete Student  
# -----  
  
elif choice == "4":  
  
    delete_name = input("Enter Student Name to Delete: ")  
  
    found = False  
  
    for student in students:  
  
        if student["Name"].lower() == delete_name.lower():  
  
            students.remove(student)  
  
            print("Student Deleted Successfully!")  
  
            found = True  
  
            break  
  
    if found == False:  
  
        print("Student Not Found.")  
  
# -----  
# Exit  
# -----  
  
elif choice == "5":  
  
    print("Thank You!")  
  
    break  
  
else:  
  
    print("Invalid Choice")
```

Output:

```
eb disinging/D1 sample/task9.py"
```

```
===== Student Record Management System =====
```

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit

Enter your choice: 1

Enter Student Name: adithyan

Enter Age: 21

Enter Branch: bca

Student Added Successfully!

```
===== Student Record Management System =====
```

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit

Enter your choice: 2

Student Records

```
{'Name': 'adithyan', 'Age': '21', 'Branch': 'bca'}
```

```
===== Student Record Management System =====
```

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit

Enter your choice: 3

Enter Student Name: adithyan

Student Found

```
{'Name': 'adithyan', 'Age': '21', 'Branch': 'bca'}
```

```
===== Student Record Management System =====
```

1. Add Student
2. Display Students

===== Student Record Management System =====

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit

Enter your choice: 4

Enter Student Name to Delete: adithyan

Student Deleted Successfully!

===== Student Record Management System =====

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit

Enter your choice: 5

Thank You!